

WASP-50b

R-0747

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_231108 · created 2026-07-28T02:27:28+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460256.72 +/- 0.096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.04 +/- 1.75 [%]
Orbital Inclination (inc)	86.24 +/- 2.88
Airmass coefficient 1 (a1)	0.56 +/- 0.32
Airmass coefficient 2 (a2)	0.34 +/- 0.28
Scatter in the residuals of the lightcurve fit is	55.92 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	7.0
Transit Duration (day)	0.042 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.1404 (deviation 0.67σ)
Duration fit vs published	0.042 d vs 0.0754 d (deviation 0.95σ)
Mid-time O–C	27.5 min
Depth SNR	3.5
Residual scatter	55.92 %

Light curve

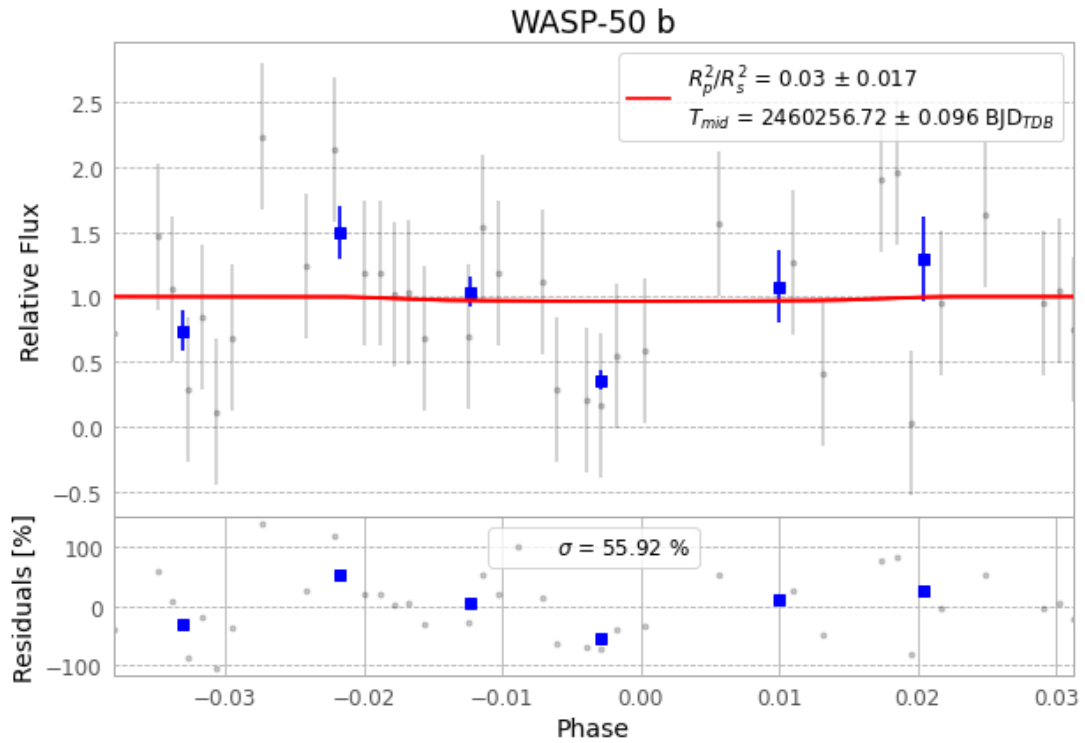


Figure 1 — FinalLightCurve_WASP-50 b_2023-11-07.png (SHA-256 9d25719d8a5dd35c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 412], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d00a686b239d11a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

66 FITS frames (43 MB), WASP-50231108032115.FITS → WASP-50231108063615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-231108.log (d48b1339e94a...), FinalLightCurve_WASP-50 b_2023-11-07.png (9d25719d8a5d...), FinalLightCurve_WASP-50 b_2023-11-07.csv (d97782c8b30b...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:27Z.